

# Vinnie Yerramadha

## AI/ML Engineer

San Francisco, CA | +1 4693193744 | [vinnieyerramadha@gmail.com](mailto:vinnieyerramadha@gmail.com) | [LinkedIn](#)

### SUMMARY

AI/ML Engineer with around 6 years of experience building machine learning and deep learning solutions across NLP, computer vision, and predictive analytics. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, and Hugging Face Transformers for model development and optimization. Skilled in MLOps practices with MLflow, Docker, Kubernetes, Airflow, and CI/CD pipelines for scalable deployment. Experienced in big data ecosystems including Spark, Kafka, and Snowflake, and cloud platforms such as AWS, GCP, and Azure. Strong background in data engineering, feature engineering, and collaboration with cross-functional teams to deliver AI-powered solutions.

### WORK EXPERIENCE

#### Shopify | USA

##### AI/ML Engineer

May 2023 – Present

- Designed and deployed real-time fraud detection pipelines using XGBoost, anomaly detection, and Kafka + Spark ML, reducing false positives by 25% and saving millions in merchant losses.
- Built a personalized recommendation engine with deep learning and reinforcement learning, improving upsell and cross-sell conversion by 18% across Shopify's marketplace.
- Developed dynamic pricing and demand forecasting models (Prophet, ARIMA, LSTMs) on millions of transactions, enabling merchants to maximize revenue while staying competitive.
- Fine-tuned LLMs (GPT, LLaMA, Hugging Face models) on merchant support data to automate Tier-1 queries, cutting manual support workload by 40% and improving response times.
- Enhanced search relevance and ranking by deploying NLP embedding models (BERT, Sentence Transformers), reducing bounce rate by 15% and increasing product discovery.
- Partnered with IT data engineers to optimize large-scale ML data pipelines on AWS and Databricks, ensuring reliable, low-latency model inference for 100M+ daily requests.
- Collaborated with business analysts and product managers to align ML outcomes with merchant needs, directly driving 12% improvement in merchant retention through churn prediction models.
- Conducted A/B testing and causal inference experiments to measure the real-world impact of ML models, ensuring that algorithmic changes improved customer experience without hurting KPIs.
- Automated MLOps workflows with MLflow, Docker, Kubernetes, and SageMaker, enabling faster iteration cycles and reducing model deployment time by 50%.
- Delivered explainable AI insights and dashboards (Tableau, Power BI) for fraud detection, personalization, and pricing, helping executives and merchants make confident, data-driven decisions.

#### NVIDIA | Bangalore, IN

##### AI/ML Engineer

Jun 2019 – Sep 2022

- Developed and deployed GPU-accelerated ML models (NLP + CV) using PyTorch Lightning, TensorFlow, and TensorRT, cutting inference latency by 35% in fraud detection and autonomous systems.
- Fine-tuned transformer models (BERT, GPT, mBERT, XLM-R) for multilingual NLP tasks such as sentiment analysis, chatbots, and regulatory text classification, boosting accuracy by 20%.
- Engineered real-time fraud detection pipelines with Kafka + Spark ML, integrating anomaly detection and graph ML algorithms to enable sub-second risk scoring across 10M+ transactions daily.
- Partnered with IT data engineering teams to design high-throughput pipelines on Snowflake, AWS, and Spark, ensuring seamless ML dataset availability with 87.7% reliability.
- Collaborated with data analysts to build customer segmentation and churn prediction models using XGBoost and clustering, driving a 15% improvement in retention strategies.
- Trained CNN and YOLO-based models for autonomous vehicle object detection, and optimized deployment on GPU edge devices (Jetson Nano + TensorRT), reducing inference time by 40%.
- Applied NLP for regulatory compliance automation, extracting entities and risk terms from SOX/GDPR documents, cutting manual review effort by 60%.
- Integrated LLMs with customer 360 data lakes using RAG (Retrieval-Augmented Generation), enabling contextual insights for segmentation, personalization, and cross-sell analytics.
- Collaborated with product managers, engineers, and business stakeholders to align AI/ML models with financial risk, fraud detection, and compliance reporting needs, saving ~\$1M in exposure.
- Streamlined MLOps lifecycle by leveraging MLflow, Airflow, Docker, and Kubernetes for experiment tracking, CI/CD, and deployment, ensuring scalable delivery of AI models in Agile squads.

### EDUCATION

#### Master's of Science in Computer

Denton, TX

University of North Texas

#### Bachelor in Computer Science

India

Malla Reddy institute of technology and sciences

## **TECHNICAL SKILLS**

**Programming Languages:** Python, SQL, Bash, C, JavaScript, HTML5, CSS3, PHP

**Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, LLaMA 2/3, GPT, CLIP, Emu, EmuEdit, EmuVideo, Hugging Face Transformers, Reinforcement Learning (RL, RLHF), Generative AI, Recommendation Systems (GEM, Andromeda)

**NLP & Conversational AI:** spaCy, BERT, NLTK, Hugging Face Transformers, Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Document Parsing, Conversational AI, OCR (Tesseract, AWS Textract)

**MLOps & Deployment:** MLflow, Docker, Kubernetes, TorchServe, Triton Inference Server, ONNX, TorchScript, Model Quantization, GitHub Actions, Jenkins, CI/CD Pipelines, Canary Deployment, Shadow Deployment

**Containerization & DevOps:** Docker, Kubernetes, GitHub Actions, Jenkins, Airflow, DVC, CI/CD Automation Databases & Big Data: Presto, Hive, Scuba, Spark SQL, Apache Spark, PySpark, Petabyte-scale Data Logs, ETL Pipelines, Distributed Systems, Scalable Data Pipelines

**Cloud Platforms:** AWS (Lambda, S3, SageMaker, CloudWatch), GCP Vertex AI, Azure ML Studio, Meta MTIA Infrastructure, NVIDIA Grace Hopper Superchip

**Responsible AI & Governance:** AI Watermarking, Brand Safety Filters, SHAP, LIME, Fairlearn, Bias Detection, GDPR/RBI/KYC/AML Compliance, Safety Filters, Human-in-the-loop Systems

**Data Visualization & Analytics:** Power BI, Tableau, Matplotlib, Seaborn, Business Intelligence Dashboards

**Frameworks & APIs:** Flask, FastAPI, RESTful APIs, ERP Integration (SAP, Oracle, NetSuite)

**Software Engineering Practices:** Agile/Scrum, A/B Testing, Software Architecture, Version Control (Git, GitLab, Bitbucket), Code Reviews, System Integration

**Tools & Environments:** VSCode, Jupyter Notebook, Git, GitLab, Bitbucket, Linux, Containerization Environments

## **PROJECTS**

### **Customer 360 AI/ML Platform | Mar 2023 – Jul 2023**

- Architected a Customer 360 AI-driven platform by integrating CRM, transactions, and web activity into an AWS S3 + Snowflake lakehouse, enabling unified ML-ready datasets.
- Built real-time streaming pipelines with Kafka + Spark Structured Streaming to capture customer interactions and feed into personalization and recommendation ML models.
- Engineered churn prediction and LTV models with XGBoost and neural nets, leveraging 100M+ customer records to achieve 92% prediction accuracy and support retention campaigns.
- Developed customer segmentation models (clustering, embedding-based) to identify high-value cohorts, boosting cross-sell opportunities by 12%.
- Designed explainable AI dashboards in Power BI to surface churn drivers, lifetime value predictors, and next-best-offer recommendations to business teams.
- Ensured responsible AI and compliance by applying GDPR-driven data governance, anonymization, and bias monitoring in ML model pipelines.

### **Regulatory Compliance AI/ML Pipeline | Aug 2023 – Dec 2023**

- Designed and implemented a compliance-focused ML pipeline centralizing audit, risk, and payments data across AWS S3, Redshift, and Snowflake for AI-powered anomaly detection.
- Built ETL + feature engineering workflows (Airflow, Python, dbt) to generate structured datasets for fraud detection and risk-scoring ML models, ensuring 99.9% data reliability.
- Trained anomaly detection models (Isolation Forest, Autoencoders, Spark ML) to flag suspicious transactions and reporting mismatches, reducing regulatory exposure by 35%.
- Applied NLP models (BERT, spaCy) on audit logs and compliance documents to automate classification, risk tagging, and entity extraction, cutting manual review time by 60%.
- Partnered with risk & audit teams to validate AI-driven outputs against compliance frameworks (SOX, GDPR, PCI DSS), reducing audit preparation time by 40%.
- Developed regulatory monitoring dashboards in Tableau to track fraud alerts, anomaly patterns, and model performance in real time.